Personal Agency and Political Agency Across the Urban-Rural Gradient

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This study uses World Values Survey 7 (2017-2023) across 47-62 countries to test urban-rural differences in personal agency and political agency. Our findings indicate that personal agency is generally no higher in urban than in rural areas, contrary to the expectation of the "city air makes you free" hypothesis–there is mostly no difference. However, political agency is significantly higher in urban areas compared to rural areas in rich countries, consistent with the "left behind ruralities" hypothesis. Furthermore, as Toennies hypothesized, those doing well financially are "active and alive" in cities or, as we find here, have greater personal agency (but not political agency). Effect sizes are considerable at about ??. As in any correlational study, these findings do not establish causality.

World Values Survey, freedom, agency, autonomy, political agency, urbanism, urban, rural, urban-rural, cities

"City life and Gesellschaft down the common people to decay and death." Ferdinand Toennies

"In the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive." Ferdinand Toennies

1 Introduction

"City air makes you free" expresses a longstanding belief that cities offer freedom from the social constraints of rural life (Yamagishi et al. 2012). Originally, referring to freedom from feudal obligations (Harvey 2012, Park et al. [1925] 1984, Yamagishi et al. 2012).[1]The saying also evokes the expectation that cities provide greater opportunities for individuals to shape their own lives.

In the social sciences, and sociology in particular, urbanization has long been viewed as one of the major drivers of human progress and empowerment. Classical sociologists described how the concentration of people in cities facilitates and necessitates a finer division of labor and greater specialization to satisfy existing and newly emerging, increasingly diverse needs (Durkheim, 2016 [1988]; Simmel, 1995 [1901–1908]). The next three paragraphs all refer to Durkheim and Simmel.

Urbanization is accompanied by and interwoven with other large-scale processes such as economic development, industrialization, and social mobility, which increase the number of social interactions and interdependencies while diversifying society and weakening collective control over the individual. Cities are a breeding ground for individual freedom–there individuals more often interact with and depend more strongly on people with different occupations, lifestyles, and values, while also enjoying greater freedom themselves to pursue different life paths in domains such as work, relationships, family, and leisure.

At the same time, the classical sociologists already warned against the relentless expansion of needs, whereby the city evokes wants and needs stimulated by ever–new products, services, and experiences, while simultaneously people become increasingly indifferent to these stimuli and to the novelty they offer. Urban life may thus expand individuals' personal agency, i.e., their freedom to shape their

[1] "Stadt Luft macht frei" (Park et al. [1925] 1984, p. 12). This saying originated in the Middle Ages, and it meant freedom from feudalism in cities; cities were non-feudal islands in a sea of feudalism (Harvey 2012).

lives in accordance with their own goals and values, but it may also constrain it by drawing them ever deeper into an endless cycle of expanding needs.

Cities may offer more choices in employment, relationships and lifestyles, but whether these possibilities translate into greater personal agency remains an empirical question.

To date, surprisingly little is known about the urban-rural divide in personal agency, and existing research has yielded mixed results (e.g. Atherton et al. 2024; Kusnali 2025; Lolle 2023). Importantly, feeling that one has control over one's own life (`personal agency`) does not necessarily mean feeling that one has a say in government decisions affecting one's community (`political agency`). Individuals may perceive themselves as capable of shaping their personal circumstances while simultaneously feeling excluded from political decision-making. Distinguishing these two dimensions is therefore essential to understanding urban-rural differences in perceived control.

Urbanization may also shape political agency, that is, individuals' confidence in their ability to influence political decisions and thereby shape their country and its future. In their classic account of the origins of modern party systems. Lipset and Rokkan (1967) identify the urban-rural cleavage as one of four fundamental social cleavages, rooted in the conflict between agrarian interests in the countryside and emerging industrial and commercial interests in the cities. Historically, this conflict found expression in the formation of parties representing rural and agrarian interests, on the one hand, and urban and industrial interests, on the other. Although the political significance of this cleavage has changed over time, the urban-rural divide remains visible in contemporary electoral politics, for instance in differences in support for Green and far-right parties (e.g., Crulli 2024; Marks et al., 2020; Rickardsson, 2021).

Although such political divides are often attributed to rural resentment, feelings of powerlessness and political inefficacy, few studies directly quantify the urban-rural gap in perceived political agency. The few that do, however, find that urban residents perceive greater political agency than their rural counterparts (e.g, García Del Horno et al. 2024, Luukkonen et al. 2022, Rowland et al. 2024).

Both personal agency and political efficacy are substantially shaped by prevailing economic conditions and individual material resources. At the societal level, economic progress can expand people's opportunities by improving infrastructure, services, and welfare provision. At the individual level, it can reduce the most existential constraints on human choice by increasing access to, and the availability of, resources that provide the objective means to exercise choice (Welzel & Inglehart, 2001; Welzel et al., 2003).

Although economic progress is ultimately expected to expand human choice and opportunities for all, urban-rural disparities in these resources and opportunities tend to be substantially wider in developing than in developed countries as has been documented, for instance for employment and wages (Ananian & Dellaferrera 2024), health care (Marthias et al. 2025), and life satisfaction (Easterlin et al. 2011). Further, urban-rural differences in outcomes are partly attributable to inequalities in individual resources such as income, as studies of right-wing voting (Rickardsson, 2021) and life satisfaction (Easterlin et al., 2011) demonstrate.

The greater range of choices available in cities does not guarantee that residents posses the means to exercise them. Thus, urbanization may expand the possibilities for personal agency without generating the same perceived freedom for everyone. This suggests that the opportunities associated with urban life are not equally accessible to all, echoing Tönnies's observation that "[...] in the city and therefore where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive" (Tönnies [1887] 2002, p. 227).

These arguments raise three related questions: Do urban residents perceive greater personal agency and political agency than rural residents? Do these differences vary with countries' levels of economic development? And does individual material well-being shape wheather urban opportunities translates into greater perceived agency? Only few studies so far have investigated urban-rural divides in personal and political agency and they have predominantly focused on Western contexts and often on a single country. Those covering multiple countries have generally not examined cross-national variation or differences between social groups. Our paper thus makes three main contributions to the literature. First, we examine urban-rural divides in personal and political agency across a more heterogeneous set of countries than previous studies and explore how these divides vary between countries. Second, we investigate whether urban-rural divides in personal agency and political agency vary systematically with countries' levels of economic development.

Third, we examine whether these divides vary according to individuals' satisfaction with material resources.

In what follows, we first review the theoretical expectations concerning urban-rural differences in personal agency and political agency. We then describe the data and analytical approach, present our results, and consider their implications and limitations.

2 Literature

2.1 Perceived Personal Agency: "City Air Makes You Free" Hypothesis

What is "perceived personal agency" or "freedom" for short?[2] To be free is to be able to choose, to control and direct one's own life. Freedom is having control–a person is free if she is her own master. It could be labeled 'psychological freedom': "Freedom means a degree of harmony between basic motives and overt behavior" (Bay 1970, p.83 cited in Okulicz-Kozaryn 2014). It is the "capability to choose," having "awareness of alternatives" and "courage to choose" (Veenhoven 2000).[3] Cities offer more choices than rural areas and, in this sense, more freedom. Yet, it is important to underscore that too much choice may also backfire and make it difficult to act or to choose, possibly leading to paralysis (Schwartz 2004).

Personal agency is a broad concept–Hitlin and Long (2009) review the concept in sociology (refer also to a recent scale (D'Italia and Okulicz-Kozaryn 2025)). Beyond its different dimensions, freedom also has value in its own right. The Universal Declaration of Human Rights recognizes freedom as a human right essential for the realization of one's human potential (art. 26, 29).

There are theoretical reasons to expect more freedom/personal agency in cities than in rural areas. In rurality, mechanical solidarity prevails–rural residents function like parts of a simple machine–groups are held together through homogeneity and similarity: similar lifestyles, roles, shared values, and morality (Durkheim [1893] 1997). Hence, rural areas enforce social conformity that is not conducive to individual freedom. Cities provide freedom from traditional rural social constraints–Gesellschaft versus Gemeinschaft–deviation from community norms is more likely to be reprimanded and ostracized in rural than urban areas. On the other hand, organic solidarity in cities allows for more differentiation and individualism. An urbanite may be more internally driven than rural residents, who are more externally constrained (Yamagishi et al. 2012). While Urbanites are more tolerant to others (Tuch 1987), they are also more reserved. Simmel (1903) observed that urbanism forces individual differentiation: in an environment characterized by diversity and constant stimulation, individuals must distinguish themselves from others through increasingly extreme means, including extravagance and exaggeration. To some extent, this pressure to stand out may foster differentiation, individualism, and personal freedom. At the same time, the overload of stimuli overwhelms individuals, prompting them to distance themselves from their environment and adopt a blasÃ© attitude toward others (Simmel 1903). Such responses may protect individuals' psyches, but emotional detachment and social withdrawal are not expressions of agency.

Cities provide more opportunities for alternative jobs and social relations, so an urbanite is freer by being less concerned about social exclusion as she has plenty of alternatives (Yamagishi et al. 2012).[4] One mechanism is higher social pressure to conform in rural areas–cities provide anonymity–indeed, concern for moral reputation is greater in rural than urban areas (Okulicz-Kozaryn 2026). The higher the population density, the lower the frequency of social comparison(Okulicz-Kozaryn 2026).

Conversely, there is also theory suggesting that people may feel less free in cities. City life is stressful, hectic, anxiety-provoking, materialistic, consumerist (Okulicz-Kozaryn 2015), and overstimulating (Simmel 1903, Lederbogen et al. 2011)–under such conditions, feeling free can be difficult. Steve Pile in his colorful writing argues that the history and past of old cities can be a weight on their inhabitants (Pile 2005a,b, Pile et al. 1999), and old cities carry melancholia (Pile 2005b). Based on the above Pile's argument, we hypothesize that the weight of history and melancholia can inhibit how free one feels.

[2]Here we focus on personal as opposed to agency/societal freedom and follow Okulicz-Kozaryn (2014) in this paragraph.

[3]The degree of perceived control over choice is called "locus of control" in psychology and Verme (2009) argues that the WVS item used here measures locus of control in addition to freedom.

[4]Also see discussion in Luca et al. (2023).

A multitude of studies have examined individual predictors of personal agency(e.g., Okulicz-Kozaryn 2014, Pitlik 2016, Nikolaev and Bennett 2016), but none of them examine urbanicity. The differences in freedom across the urban-rural gradient have been examined in only a handful of studies covering very few countries. These studies offer a mixed picture, examining both general perceptions of agency and control over specific aspects of daily life. Lolle (2023) finds no differences in agency across communities of different sizes in Denmark. Atherton et al. (2024) find that psychological well-being–which entails agency as a subcomponent–is lower in the most rural places compared to the most urban places in the USA; however, these effects become statistically insignificant once socio-demographic factors are controlled for. Kusnali et al. (2025) find that women in Indonesia have more decision making agency in many (i.e., household expenditures, food expenditures, time allocation to social activities) but not all (e.g., contraceptive use) aspects of daily life in urban compared to rural areas. Indian women have more autonomy in cities than in rural areas, both in personal and household decisions (Mishra 2014). Båge et al. find no urban-rural difference in people's perceived reproductive agency, i.e. how much freedom of choice and control they have over family planning, in Ethiopia, Kenya, and Zimbabwe (Båge et al. 2024) and sub-Saharan Africa (Båge et al. 2026). Taken together, these findings leave open the question of whether urban-rural differences in perceived personal agency follow an universal pattern acrross countries.

2.2 Perceived Political Agency: "Left Behind Ruralities" Hypothesis

As the world continues to urbanize, rural areas are disappearing (Pospěch et al. 2024). Rurality in the developed world is increasingly peripheral, loosing power and voice relative to urban centers and being abandoned and disrespected–thereby fueling rural discontent, resentment, and shame (see superb accounts in Hanson (2015), Fuller (2017)). It is therefore hardly surprising that the more urbanized affluent democracies are increasingly witnessing a growing political divide between urban and rural areas, evident in voting patterns (Huijsmans and Rodden 2025, Mettler and Brown 2022), political trust (Mitsch et al. 2021), and perceptions of individual and community recognition (Hyland et al. 2024).

There is rural resentment, e.g., people feel that policy is imposed on them, rather than developed with or for them; and there is racial resentment where rural Whites resent urban minorities and immigrants getting better treatment than them (Diamond 2023). Although rural life involves reliance on community and tight connections, self-reliance, autonomy, and freedom also remain core values (Diamond 2023).

Rural areas are 'left-behind' (Ulrich-Schad and Duncan 2021), 'places that don't matter' (Rodríguez-Pose 2018)—they are second-tier areas with limited or deteriorating opportunities. When people's opportunities are constrained, their freedom to exercise agency is likewise restricted (Ryan and Deci 2004, Welzel 2013). Exercising agency requires access to real and substantively different opportunities from which individuals can choose in pursuing lives that are consistent with their own goals and values (Sen 1993, 1999, 2005). Limited opportunities may therefore undermine both people's perceived ability to shape their own lives in accordance with these goals and values, and their perceived ability to shape the conditions in which their lives are embedded, such as institutions, infrastructure, and the built and natural environment. We therefore expect people living in 'left-behind' rural areas to perceive themselves as having less influence over the conditions in which they are embedded and that shape their lives, and consequently to perceive lower levels of political agency.

Only a few studies have investigated urban-rural differences in political agency. Studies investigating the urban-rural divide in political agency consistently find that people in urban areas perceive themselves as more politically efficacious than those in rural areas, both in Europe (García Del Horno et al. 2024, Luukkonen et al. 2022) and the United States (Rowland et al. 2024).

Exploring rural resentment in Europe and the United States, Margalit et al. (2025) find that rural residents who feel they have little or no say in what the government does are more likely to vote for populist parties in Europe and for Trump in the United States. Political agency has been shown to be positively linked to political engagement (Accetta 2023). Self-efficacy and political efficacy have been shown to have opposing effects on the justification of terrorism and political violence: people who perceive themselves as highly

self-efficacious and those who feel they have little political efficacy are less likely to justify such actions (Julkif 2022).

Overall, these studies suggest that rural residents may feel less politicall efficacy/agency than urban resident. But no one has investigated whether this relationship is universal across different contexts (such as countries with high or low GDP). Wether these urban-rural differences are universal across countries with different economic development remains an open question.

3 Hypotheses

Against this backdrop, we investigate the following hypotheses:
H1: Respondents residing in urban areas report significantly higher levels of personal and political agency than those residing in rural areas.

Given that economic and social progress unfolds unevenly across urban and rural areas depending on a country's level of development, we further hypothesize that the urban-rural divide is more pronounced in less developed countries:
H2: The urban-rural divide in personal and political agency is significantly larger in low- and middle-income countries than in high-income countries.

Finally, as differences in individual material resources may account for part of the urban-rural divide, we examine whether its magnitude varies across levels of individual material resources:
H3: The urban-rural divide in personal and political agency is smaller at higher levels of individual material resources.
A variation on this hypothesis is following the Toennies quote:
H3a: The "more" of agency in cities mostly stems from well-to-do urbanites.

4 Data and analytical strategy

To test these hypotheses, we use data from Wave 7 of the World Values Survey (WVS; Haerpfer et al., 2022). The WVS is a repeated cross-sectional survey conducted approximately every five years since 1981 to study social, political, economic, religious, and cultural values, beliefs, and norms worldwide (WVS 2026a, 2026b). It draws representative samples of each country's adult population using simple or multistage random sampling, depending on the country (WVS 2026c). Data for Wave 7 were collected between 2017 and 2023, primarily through face-to-face interviews, although other modes of data collection, such as telephone interviews and self-completion, were also used in some countries (WVS 2026c).

WVS World Value Survey (2026a): Data and Documentation, Data Download, Documentation for Download. `https://www.worldvaluessurvey.org/WVSContents.jsp`, last accessed: 10.09.2026. WVS World Value Survey (2026b): Who we are, Funding Partners. `https://www.worldvaluessurvey.org/WVSContents.jsp`, last accessed: 10.09.2026. WVS World Value Survey (2026c): WVS Wave 7 (2017-2022). `https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp`, last accessed: 10.09.2026.

4.1 Operationalization

Urban-Rural. Although strictly speaking, urban-rural is a gradient rather than a dichotomy (Okulicz-Kozaryn 2017), we employ a categorical operationalization of the degree of urbanization. This approach is consistent with the theoretical assumptions outlined above, which commonly postulate a rift between the rural countryside on the one hand and the metropolis on the other. Additionally, as survey respondents are often unevenly distributed across the community-size categories captured in surveys, collapsing these categories is common practice in empirical comparative urban-rural research (Okulicz-Kozaryn and Valente 2021). The original town-size variable, coded by interviewers into eight categories ranging from fewer than 2,000 to 500,000 or more inhabitants, was thus collapsed into

three categories: (1) rural areas, comprising communities with fewer than 20,000 inhabitants; (2) towns, comprising communities with 20,000 to fewer than 500,000 inhabitants; and (3) cities, comprising communities with 500,000 or more inhabitants.

In the appendix we do several robustness checks, including dropping countries with fewer than 50 observations per category extreme category. Results are similar except for interactions with financial satisfactuoin where we drop if <20k <50obs (results are robust when droping countries where >500k <50obs).

Personal agency. We operationalize personal agency using respondents' self-assessed freedom of choice and control, measured with the following item: "Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'no choice at all' and 10 means 'a great deal of choice' to indicate how much freedom of choice and control you feel you have over the way your life turns out." Higher values indicate stronger perceived personal agency. This item is commonly used to capture individuals' perceived agency (e.g., Verme 2009, Inglehart et al. 2008, Veenhoven 2000, Okulicz-Kozaryn 2014) and related concepts such as autonomy, locus of control, and fatalism (see Kouba (2016) for a conceptual discussion and Steckermeier 2022 for a review of empirical research based on this item).

All variables including controls are listed in the appendix and there are also shown distributions of all the variables. We are controlling for married, employment type dummies (full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction,[5] health, kids, and importance of religion.

Leonie TODO: here lit for personal agency predictors

Additional robustness test were conducted resulting in oversaturated models; the addition of these controls yield larger coefficients, although overall results are the same. Refer to the Appendix for more detail.

Political agency. We operationalize political agency via respondents' answers to the question: "How much would you say the political system in your country allows people like you to have a say in what the government does?" Responses range from 1 ("A great deal") to 5 ("Not at all"). We reverse the response categories so that higher values indicate stronger perceived political agency. The item is regularly used to capture external political efficacy (e.g., Sumino, 2025; Bienstman et al., 2023) and related concepts such as having a political voice (Prats et al., 2023) and perceptions of political unfairness (Im, 2024). Unfortunately, this item was not fielded in Bolivia, Chile, China, Germany, Ecuador, Greece, South Korea, Puerto Rico, Romania, Russia, Serbia, Thailand, Turkey, and the United States. The sample for this part of the analysis is thus reduced to XX countries (from XX for analyses with personal agency).

@Adam: add the number of countries

Control variables. In all analyses pertaining to political agency we control for the following factors which have been shown to be associated with political agency as well as to differ between urban and rural areas: education (political agency: (Emmenegger et al. 2015, García Del Horno et al. 2024, Marx and Nguyen 2016, Rowland et al. 2024) urban-rural: (Van Maarseveen 2021, Zahl-Thanem and Rye 2024), employment (political agency: Emmenegger et al. (2015), Luukkonen et al. (2022), Marx and Nguyen (2016) urban-rural: Eurostat (2026), Ananian and Dellaferrera (2024), material resources (political agency: Emmenegger et al. (2015), Luukkonen et al. (2022), Marx and Nguyen (2016), Rowland et al. (2024) ; urban-rural: Finnemann et al. (2024), Kpatcha et al. (2026), Meloni et al. (2024), health (political agency: García Del Horno et al. (2024), Gidengil and Wass (2024), Shore et al. (2019); urban-rural: MacKinnon et al. (2023), Menashe-Oren and Masquelier (2022), and religiosity (political agency: Marx and Nguyen (2016), Rowland et al. (2024) ; urban-rural: Balazka (2020), Meulemann and Schmidt-Catran (2020). In all analyses pertaining to personal agency we additionally control for having a family because living with a partner and/or child/ren has been shown to be associated with personal agency (e.g., Nikolaev and Bennett (2016) Steckermeier 2024; Symoens et al., 2014) as well as to differ between urban and rural areas (e.g., Gautier et al., 2010; Stockdale and Catney (2014).

[5]Financial satisfaction is key–not everyone needs the same amount of money to be financially satisfied and feel free. This is measured as "How satisfied are you with the financial situation of your household?"

5 Results

5.1 Interactions with Level of Development

We begin by reporting in table 1 mean personal agency and political agency by settlement size and level of development. It is important to underscore that even finding no difference in personal agency level between urban and rural areas would be worth reporting, given our expectation of an urban advantage and positive relationship.

Table 1: Means by level of devlopment. Categorization of development according to World Bank income groups. LIC/LMIC= Lower income countries and lower middle income countries. UMIC = Upper middle income countries. HIC = High income countries.

	personal agency				**political agency**			
	country development:				country development:			
place	overall	low/lower middle	upper middle	high	overall	low/lower middle	upper middle	high
<20k	7.20	7.11	7.26	7.24	2.96	3.03	3.06	2.67
20-500k	7.27	7.21	7.21	7.36	2.88	2.95	2.83	2.87
>500k	7.12	6.96	7.40	6.98	2.89	2.67	2.86	2.99

Table 1 shows the average reported levels of personal and political agency across rural, town, and city areas, by societal level of development and individual-level material resources.

Overall, personal agency is lowest in cities and highest in towns, with rural areas in between. This pattern is also observed in less developed countries and especially high development countries, but not in upper middle, where the ranking is reversed–it has highest levels in cities. Overall, however, these differences are small and disappear after controlling for predictors of freedom.

Political agency, by contrast, is highest in rural areas, followed by cities and towns, which show practically identical levels. However, patterns of political agency differ considerably by a country's level of development. In less developed countries, average perceived political agency is highest in rural areas and lowest in cities. In in high developed countries, this pattern is reversed: people in cities report higher political agency than those in towns, who in turn report higher agency than those in rural areas. In upper middle, the pattern largely mirrors the overall results, with political agency highest in rural areas, followed by cities and then towns, where levels are relatively similar.

And that last difference holds even when controlling for predictors of political agency. Margins are plotted in figure 1. While the confidence intervals are wide, the interaction between >500k and high developed countries is significant versus the base case of <20k and low/lower development countries. The results for personal agency are insignificant.

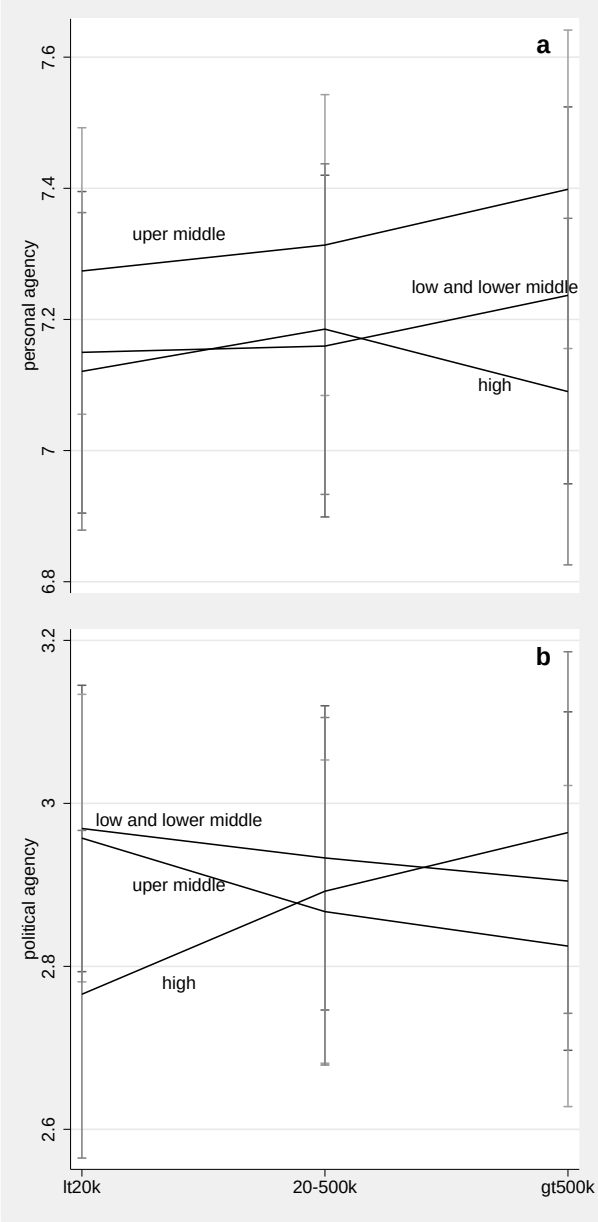


Figure 1: m-fps these results are robust when droping countries where >500k <50obs and <20k <50obs

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5.2 Interactions with Financial Satisfaction

In table 2 `personal agency` for `bottom 4` categories of `financial satisfaction`: highest in <20k, and lowest in >500k. `personal agency` for `middle 3` categories of `financial satisfaction`: highest in 20-500k; and likewise for `top 3`.

`Political agency` for `bottom 4` categories of `financial satisfaction`: lowest in the middle 20-500k. `Political agency` for `middle 3` categories of `financial satisfaction`: highest in the smallest places <20k; and likewise for the `top 3`.

Table 2: Means by level of financial satisfaction.

	personal agency				**political agency**			
	financial satisfaction:				financial satisfaction:			
place	overall	bottom 4	middle 3	top 3	overall	bottom 4	middle 3	top 3
<20k	7.20	6.25	7.04	8.09	2.96	2.78	2.96	3.11
20-500k	7.27	6.15	7.15	8.12	2.88	2.66	2.87	3.01
>500k	7.12	6.10	6.99	8.06	2.89	2.72	2.89	3.02

The interactions with financial satisfaction results for both personal agency and political agency are visualized in figure 2. We focus here on visual interpretations using marginsplot, but all underlying regressions are in the appendix. Personal agency results in panel a are significant. The relevant significant interactions we report are for the smallest places $< 20k$ (against the base case of $> 500k$) with top category(ies).

In panel a, personal agency increases with size among respondents in the top 3 and also middle 3 financial-satisfaction categories from $< 20k$ to $> 500k$, but decreases for the bottom 3 categories. In other words, personal agency differences are magnified in the largest cities across financial satisfaction groups–financially satisfied respondents report a larger advantage in freedom over financially dissatisfied respondents in cities. The corresponding interaction results for political agency in panel b are insignificant.

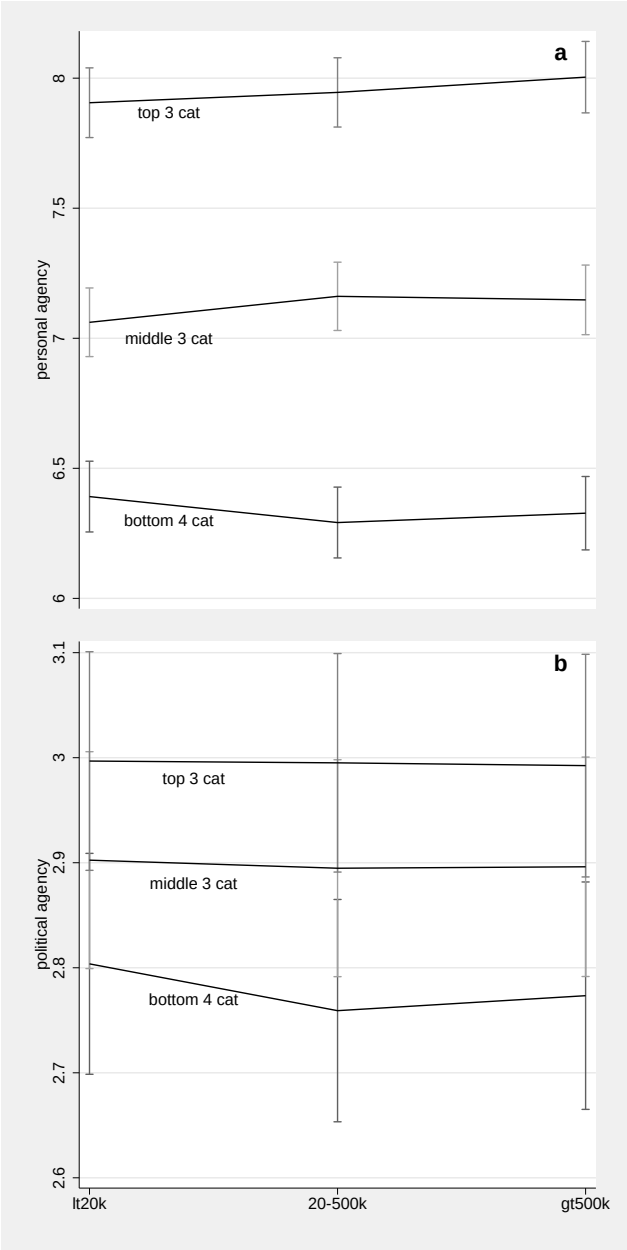


Figure 2: zz these results are robust when droping countries where >500k <50obs; but not where <20k <50obs

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6 Discussion and Conclusion

The expectation that cities provide greater freedom is plausible, but in our pooled sample without interactions, we found no significant differences when controlling for country dummies. It is worth highlighting that finding no urban-rural effect for personal agency in most countries is a finding worth reporting–the expectation from theory and from the literature is that the bigger places have more freedom.

For political agency, however, we find it to be significantly higher in urban compared to rural areas in rich countries as hypothetized in the "left behind ruralities" hypothesis. We find support for Toennies' idea that: "in the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive" (Tönnies [1887] 2002, p. 227). There are significant interactions in the pooled sample for financial satisfaction.

Concurrent with results from Okulicz-Kozaryn (2024), urbanites probably have, in some ways, fuller and freer lives, but perhaps they realize more what they miss out relative to other urbanites, and so they feel relatively less free. It's possible that the availability of opportunities does not necessarily translate into the ability to use them. Costs, stress, unequal resources may limit one's choices even when possibilities are endless.

Another mechanism could be that people tend to compare themselves with those who are better off(Frey and Stutzer 2002) and there are more people to compare to in urban areas. The inequality ladder is taller and the pond is bigger–a small fish in a big pond (city) seems to be less free than a big fish in a small pond (rural area).

In poorer/less economically developed countries, there are more differences in freedom across the rural-urban gradient, mostly less freedom in smaller places[6]–this makes sense–smaller places in developing countries can be very traditional and very poor–freedom is difficult in such places. In richer countries, on the other hand, one can already be free just about anywhere, there is no need for a large city. This is a similar to findings in the literature with respect to happiness (Okulicz-Kozaryn 2015)–only in the very poorest places of all such as Sub-Saharan Africa, is happiness greater in larger cities as life is unbearable and lacks necessities such as shelter and clean water in rural areas. Everywhere else urbanites tend to be less happy (Okulicz-Kozaryn and Valente 2021).

Overall, our findings suggest that the freedom associated with urban areas depends on which aspect we consider. Cities offer more opportunities and choices, but these do not necessarily translate into greater perceived freedom and control over one's life. Yet, for political agency, we find an urban advantage in rich countries, consistent with the "left behind ruralities" hypothesis. In rich countries, rural residents report lower political agency, even though personal agency generally shows little urban-rural difference. Hence, they may feel that they have control over their own lives while feeling that they have less say in government decisions. Future research could shed light on our findings by investigating whether rural residents' perception of having less political voice reflect differences in institutional response, political representation, or access to decision making.

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[6](But sometimes more, refer to appendix)

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ONLINE APPENDIX (SOM)

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead as SOM 'Supplementary Online Material']

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1 Variables' Definitions and Descriptive Statistics

TODO aok var des variable definitions table here

hist histograms here

2 Regressions underlying body results

2.1 Interactions with level of development

```
 mixed polSay i.t3##i.iii  i.X025R i.emp satFin health  rel_imp  || cc: i.t3  ,mle //final! yes educ; no mar no kids
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -101253.19
Iteration 1:  Log likelihood = -101253.19
Computing standard errors ...
Mixed-effects ML regression                     Number of obs    =  64,862
Group variable: cc                              Number of groups =      48
                                                Obs per group:
                                                              min =     726
                                                              avg = 1,351.3
                                                              max =   3,997
                                                Wald chi2(20)    =  890.11
Log likelihood = -101253.19                     Prob > chi2      =  0.0000
------------------------------------------------------------------------------
      polSay | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
          t3 |
    20-500k  |      -0.04       0.04    -0.96    0.34        -0.11        0.04
     gt500k  |      -0.06       0.06    -1.07    0.28        -0.18        0.05
             |
         iii |
           2 |      -0.01       0.13    -0.09    0.93        -0.26        0.24
           3 |      -0.20       0.14    -1.49    0.14        -0.47        0.06
             |
       t3#iii |
  20-500k#2  |      -0.05       0.05    -1.02    0.31        -0.16        0.05
  20-500k#3  |       0.16       0.06     2.73    0.01         0.05        0.28
   gt500k#2  |      -0.07       0.08    -0.87    0.39        -0.22        0.09
   gt500k#3  |       0.26       0.09     2.87    0.00         0.08        0.44
             |
        X025R |
     Middle  |       0.01       0.01     0.50    0.62        -0.02        0.03
      Upper  |       0.05       0.01     3.78    0.00         0.02        0.07
             |
         emp |
  Part time  |       0.00       0.02     0.04    0.97        -0.03        0.03
Self employed|      -0.02       0.01    -1.58    0.11        -0.05        0.01
     Retired |       0.01       0.02     0.37    0.71        -0.03        0.04
   Housewife |      -0.07       0.02    -4.40    0.00        -0.10       -0.04
    Students |       0.00       0.02     0.19    0.85        -0.04        0.04
 Unemployed  |      -0.06       0.02    -3.09    0.00        -0.10       -0.02
       Other |      -0.03       0.05    -0.72    0.47        -0.13        0.06
             |
      satFin |       0.03       0.00    17.07    0.00         0.03        0.04
      health |       0.07       0.01    12.53    0.00         0.06        0.08
     rel_imp |       0.07       0.01    12.29    0.00         0.06        0.08
       _cons |       2.27       0.09    23.94    0.00         2.09        2.46
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Independent              |
                  var(2.t3) |       0.02       0.00          0.01        0.03
                  var(3.t3) |       0.03       0.01          0.02        0.06
                 var(_cons) |       0.13       0.03          0.09        0.20
-----------------------------+------------------------------------------------
              var(Residual) |       1.32       0.01          1.31        1.34
------------------------------------------------------------------------------
LR test vs. linear model: chi2(3) = 4821.97               Prob > chi2 = 0.0000
Note: LR test is conservative and provided only for reference.
.
. margins,  at(t3=(1 2 3)  iii=(1 2 3))
Predictive margins                                  Number of obs = 64,862
Expression: Linear prediction, fixed portion, predict()
1._at: t3  = 1
       iii = 1
2._at: t3  = 1
       iii = 2
3._at: t3  = 1
       iii = 3
4._at: t3  = 2
       iii = 1
5._at: t3  = 2
       iii = 2
6._at: t3  = 2
       iii = 3
7._at: t3  = 3
       iii = 1
8._at: t3  = 3
       iii = 2
9._at: t3  = 3
       iii = 3
```

```
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |       2.97       0.09    33.11    0.00         2.79        3.14
          2  |       2.96       0.09    32.86    0.00         2.78        3.13
          3  |       2.77       0.10    26.97    0.00         2.56        2.97
          4  |       2.93       0.10    30.80    0.00         2.75        3.12
          5  |       2.87       0.09    30.20    0.00         2.68        3.05
          6  |       2.89       0.11    26.58    0.00         2.68        3.11
          7  |       2.90       0.11    27.41    0.00         2.70        3.11
          8  |       2.82       0.10    28.10    0.00         2.63        3.02
          9  |       2.96       0.11    26.19    0.00         2.74        3.19
------------------------------------------------------------------------------

. marginsplot,saving(a1,replace)ytitle("political say")scheme(s2mono)title("") text(2.98 1.4 "low and lower middle")text(2.9 1.4 "uper middle")
> text(2.78 1.4 "high")recast(line) legend(off)xtitle("size of a place") //xsc(off)
Variables that uniquely identify margins: t3 iii
file a1.gph saved

. mixed free i.t3##i.iii  mar i.emp satFin health kids rel_imp  || cc:  i.t3 ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -181550.65
Iteration 1:  Log likelihood = -181550.65
Computing standard errors ...
Mixed-effects ML regression                     Number of obs    =   84,775
Group variable: cc                              Number of groups =       61
                                                Obs per group:
                                                              min =      849
                                                              avg =  1,389.8
                                                              max =    4,018
                                                Wald chi2(20)    = 10564.59
Log likelihood = -181550.65                     Prob > chi2      =   0.0000
------------------------------------------------------------------------------
        free | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
          t3 |
    20-500k  |       0.01       0.06     0.17    0.86        -0.10        0.12
     gt500k  |       0.09       0.08     1.04    0.30        -0.08        0.25
             |
         iii |
           2 |       0.12       0.17     0.74    0.46        -0.20        0.45
           3 |      -0.03       0.18    -0.16    0.87        -0.37        0.32
             |
      t3#iii |
20-500k#2    |       0.03       0.07     0.40    0.69        -0.12        0.18
20-500k#3    |       0.05       0.08     0.67    0.50        -0.10        0.21
 gt500k#2    |       0.04       0.11     0.36    0.72        -0.17        0.24
 gt500k#3    |      -0.12       0.12    -1.00    0.32        -0.35        0.11
             |
         mar |       0.02       0.02     1.38    0.17        -0.01        0.06
             |
         emp |
   Part time |      -0.09       0.03    -3.31    0.00        -0.15       -0.04
Self employed|      -0.01       0.02    -0.44    0.66        -0.06        0.04
     Retired |      -0.06       0.02    -2.33    0.02        -0.11       -0.01
   Housewife |      -0.20       0.02    -8.49    0.00        -0.25       -0.16
    Students |      -0.21       0.03    -6.12    0.00        -0.27       -0.14
  Unemployed |      -0.11       0.03    -3.71    0.00        -0.17       -0.05
       Other |      -0.17       0.07    -2.49    0.01        -0.31       -0.04
             |
      satFin |       0.26       0.00    79.88    0.00         0.25        0.26
      health |       0.34       0.01    38.79    0.00         0.33        0.36
        kids |       0.03       0.01     4.58    0.00         0.01        0.04
     rel_imp |       0.03       0.01     3.81    0.00         0.02        0.05
       _cons |       4.16       0.13    30.94    0.00         3.90        4.42
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Independent              |
                   var(2.t3) |       0.03       0.01          0.02        0.06
                   var(3.t3) |       0.06       0.02          0.03        0.11
                  var(_cons) |       0.27       0.05          0.19        0.39
-----------------------------+------------------------------------------------
               var(Residual) |       4.22       0.02          4.18        4.26
------------------------------------------------------------------------------
LR test vs. linear model: chi2(3) = 4153.75               Prob > chi2 = 0.0000
Note: LR test is conservative and provided only for reference.

. margins,  at(t3=(1 2 3)  iii=(1 2 3))
Predictive margins                                  Number of obs = 84,775
Expression: Linear prediction, fixed portion, predict()
1._at: t3  = 1
       iii = 1
2._at: t3  = 1
       iii = 2
3._at: t3  = 1
       iii = 3
4._at: t3  = 2
       iii = 1
5._at: t3  = 2
       iii = 2
6._at: t3  = 2
       iii = 3
7._at: t3  = 3
       iii = 1
8._at: t3  = 3
       iii = 2
9._at: t3  = 3
       iii = 3
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |       7.15       0.13    57.16    0.00         6.90        7.40
          2  |       7.27       0.11    65.26    0.00         7.06        7.49
```

```
          3  |       7.12       0.12    57.64    0.00         6.88        7.36
          4  |       7.16       0.13    53.85    0.00         6.90        7.42
          5  |       7.31       0.12    62.48    0.00         7.08        7.54
          6  |       7.19       0.13    55.87    0.00         6.93        7.44
          7  |       7.24       0.15    49.35    0.00         6.95        7.52
          8  |       7.40       0.12    59.72    0.00         7.16        7.64
          9  |       7.09       0.13    52.58    0.00         6.83        7.35
------------------------------------------------------------------------------
. marginsplot,saving(a2,replace)ytitle("freedom")scheme(s2mono)title("") text(7.25 2.7 "low and lower middle")text(7.33 1.4 "uper middle")text(
> 7.1 2.7 "high")recast(line) legend(off)xtitle("size of a place") xsc(off)
Variables that uniquely identify margins: t3 iii
file a2.gph saved
graph combine a2.gph a1.gph, cols(1)  xsize(2) ysize(4) scale(1.1)
gr export m-fps.pdf,replace
```

.19

2.1.1 which differences are significant at what level

`Personal Agency`: With one exception, there are no significant differences in the level of perceived agency between rural, town, and city areas irrespective of the level of development: In UMIC, people in city areas report significantly more personal agency than in rural areas (b=0.12*).

`Political Agency`: In LIC/LMIC there is no significant difference in political agency between rural, town and city areas. In UMIC rural residents report significantly stronger perceived political agency than town (b=0.09*) and city (b=0.13**) residents - whereas political agency does not differ significantly between town and city areas. In HIC people in rural areas report significantly less political agency than people in town (b=-0.13**) and city (-0.20***) areas; there is no significant difference in the level of perceived political agency between town and city areas.

```
//this is for free
FIRST JUST ONLY RELEVANT ONES, and below that all
------------------------------------------------------------------------------------------
                             |            Delta-method    Unadjusted           Unadjusted
                             |   Contrast   std. err.      z    P>|z|     [95% conf. interval]
-----------------------------+------------------------------------------------------------
                      _at#t3 |
  (1#20-500k) vs (1#lt20k)   |       0.01       0.06     0.17    0.86        -0.10        0.12
   (1#gt500k) vs (1#lt20k)   |       0.09       0.08     1.04    0.30        -0.08        0.25
 (1#gt500k) vs (1#20-500k)   |       0.08       0.09     0.82    0.41        -0.11        0.26
  (2#20-500k) vs (2#lt20k)   |       0.04       0.05     0.80    0.42        -0.06        0.14
   (2#gt500k) vs (2#lt20k)   |       0.12       0.06     1.96    0.05        -0.00        0.25
 (2#gt500k) vs (2#20-500k)   |       0.08       0.07     1.16    0.25        -0.06        0.23
  (3#20-500k) vs (3#lt20k)   |       0.06       0.06     1.08    0.28        -0.05        0.18
   (3#gt500k) vs (3#lt20k)   |      -0.03       0.08    -0.38    0.71        -0.19        0.13
 (3#gt500k) vs (3#20-500k)   |      -0.10       0.09    -1.08    0.28        -0.27        0.08
------------------------------------------------------------------------------------------

 //this is for free
. margins t3, at(iii=(1 2 3)) pwcompare(effects)
Pairwise comparisons of predictive margins              Number of obs = 84,775
Expression: Linear prediction, fixed portion, predict()
1._at: iii = 1
2._at: iii = 2
3._at: iii = 3
------------------------------------------------------------------------------------------
                             |            Delta-method    Unadjusted           Unadjusted
                             |   Contrast   std. err.      z    P>|z|     [95% conf. interval]
-----------------------------+------------------------------------------------------------
                      _at#t3 |
  (1#20-500k) vs (1#lt20k)   |       0.01       0.06     0.17    0.86        -0.10        0.12
   (1#gt500k) vs (1#lt20k)   |       0.09       0.08     1.04    0.30        -0.08        0.25
    (2#lt20k) vs (1#lt20k)   |       0.12       0.17     0.74    0.46        -0.20        0.45
  (2#20-500k) vs (1#lt20k)   |       0.16       0.17     0.96    0.34        -0.17        0.50
   (2#gt500k) vs (1#lt20k)   |       0.25       0.18     1.41    0.16        -0.10        0.59
    (3#lt20k) vs (1#lt20k)   |      -0.03       0.18    -0.16    0.87        -0.37        0.32
  (3#20-500k) vs (1#lt20k)   |       0.04       0.18     0.20    0.84        -0.32        0.39
   (3#gt500k) vs (1#lt20k)   |      -0.06       0.18    -0.33    0.75        -0.42        0.30
 (1#gt500k) vs (1#20-500k)   |       0.08       0.09     0.82    0.41        -0.11        0.26
  (2#lt20k) vs (1#20-500k)   |       0.11       0.17     0.66    0.51        -0.23        0.45
(2#20-500k) vs (1#20-500k)   |       0.15       0.18     0.87    0.38        -0.19        0.50
 (2#gt500k) vs (1#20-500k)   |       0.24       0.18     1.32    0.19        -0.12        0.60
  (3#lt20k) vs (1#20-500k)   |      -0.04       0.18    -0.21    0.83        -0.39        0.32
(3#20-500k) vs (1#20-500k)   |       0.03       0.19     0.14    0.89        -0.34        0.39
 (3#gt500k) vs (1#20-500k)   |      -0.07       0.19    -0.37    0.71        -0.44        0.30
   (2#lt20k) vs (1#gt500k)   |       0.04       0.18     0.20    0.84        -0.32        0.40
 (2#20-500k) vs (1#gt500k)   |       0.08       0.19     0.41    0.68        -0.29        0.44
  (2#gt500k) vs (1#gt500k)   |       0.16       0.19     0.84    0.40        -0.21        0.54
   (3#lt20k) vs (1#gt500k)   |      -0.12       0.19    -0.60    0.55        -0.49        0.26
 (3#20-500k) vs (1#gt500k)   |      -0.05       0.20    -0.26    0.79        -0.43        0.33
  (3#gt500k) vs (1#gt500k)   |      -0.15       0.20    -0.74    0.46        -0.54        0.24
  (2#20-500k) vs (2#lt20k)   |       0.04       0.05     0.80    0.42        -0.06        0.14
   (2#gt500k) vs (2#lt20k)   |       0.12       0.06     1.96    0.05        -0.00        0.25
    (3#lt20k) vs (2#lt20k)   |      -0.15       0.17    -0.92    0.36        -0.48        0.17
  (3#20-500k) vs (2#lt20k)   |      -0.09       0.17    -0.52    0.60        -0.42        0.25
   (3#gt500k) vs (2#lt20k)   |      -0.18       0.18    -1.05    0.29        -0.53        0.16
 (2#gt500k) vs (2#20-500k)   |       0.08       0.07     1.16    0.25        -0.06        0.23
  (3#lt20k) vs (2#20-500k)   |      -0.19       0.17    -1.13    0.26        -0.53        0.14
(3#20-500k) vs (2#20-500k)   |      -0.13       0.17    -0.74    0.46        -0.47        0.21
 (3#gt500k) vs (2#20-500k)   |      -0.22       0.18    -1.25    0.21        -0.57        0.13
```

```
   (3#lt20k) vs (2#gt500k)  |      -0.28       0.18    -1.59    0.11        -0.62        0.07
 (3#20-500k) vs (2#gt500k)  |      -0.21       0.18    -1.19    0.23        -0.56        0.14
  (3#gt500k) vs (2#gt500k)  |      -0.31       0.18    -1.68    0.09        -0.67        0.05
  (3#20-500k) vs (3#lt20k)  |       0.06       0.06     1.08    0.28        -0.05        0.18
   (3#gt500k) vs (3#lt20k)  |      -0.03       0.08    -0.38    0.71        -0.19        0.13
 (3#gt500k) vs (3#20-500k)  |      -0.10       0.09    -1.08    0.28        -0.27        0.08
----------------------------------------------------------------------------------------------

//this is for PolSay
FIRST JUST ONLY RELEVANT ONES, and below that all
----------------------------------------------------------------------------------------------
                           |            Delta-method    Unadjusted           Unadjusted
                           |   Contrast   std. err.       z    P>|z|     [95% conf. interval]
---------------------------+------------------------------------------------------------------
                    _at#t3 |
  (1#20-500k) vs (1#lt20k) |      -0.04       0.04    -0.96    0.34        -0.11        0.04
   (1#gt500k) vs (1#lt20k) |      -0.06       0.06    -1.07    0.28        -0.18        0.05
 (1#gt500k) vs (1#20-500k) |      -0.03       0.07    -0.42    0.67        -0.16        0.10
  (2#20-500k) vs (2#lt20k) |      -0.09       0.04    -2.41    0.02        -0.16       -0.02
   (2#gt500k) vs (2#lt20k) |      -0.13       0.05    -2.63    0.01        -0.23       -0.03
 (2#gt500k) vs (2#20-500k) |      -0.04       0.06    -0.72    0.47        -0.16        0.07
  (3#20-500k) vs (3#lt20k) |       0.13       0.05     2.74    0.01         0.04        0.22
   (3#gt500k) vs (3#lt20k) |       0.20       0.07     2.88    0.00         0.06        0.33
 (3#gt500k) vs (3#20-500k) |       0.07       0.08     0.93    0.35        -0.08        0.22
----------------------------------------------------------------------------------------------

//this is for PolSay
 margins t3, at(iii=(1 2 3)) pwcompare(effects)
Pairwise comparisons of predictive margins             Number of obs = 64,862
Expression: Linear prediction, fixed portion, predict()
1._at: iii = 1
2._at: iii = 2
3._at: iii = 3
----------------------------------------------------------------------------------------------
                           |            Delta-method    Unadjusted           Unadjusted
                           |   Contrast   std. err.       z    P>|z|     [95% conf. interval]
---------------------------+------------------------------------------------------------------
                    _at#t3 |
  (1#20-500k) vs (1#lt20k) |      -0.04       0.04    -0.96    0.34        -0.11        0.04
   (1#gt500k) vs (1#lt20k) |      -0.06       0.06    -1.07    0.28        -0.18        0.05
    (2#lt20k) vs (1#lt20k) |      -0.01       0.13    -0.09    0.93        -0.26        0.24
  (2#20-500k) vs (1#lt20k) |      -0.10       0.13    -0.78    0.43        -0.36        0.15
   (2#gt500k) vs (1#lt20k) |      -0.14       0.13    -1.07    0.28        -0.41        0.12
    (3#lt20k) vs (1#lt20k) |      -0.20       0.14    -1.49    0.14        -0.47        0.06
  (3#20-500k) vs (1#lt20k) |      -0.08       0.14    -0.55    0.59        -0.35        0.20
   (3#gt500k) vs (1#lt20k) |      -0.01       0.14    -0.03    0.97        -0.29        0.28
 (1#gt500k) vs (1#20-500k) |      -0.03       0.07    -0.42    0.67        -0.16        0.10
  (2#lt20k) vs (1#20-500k) |       0.02       0.13     0.19    0.85        -0.23        0.28
(2#20-500k) vs (1#20-500k) |      -0.07       0.13    -0.49    0.62        -0.33        0.20
 (2#gt500k) vs (1#20-500k) |      -0.11       0.14    -0.78    0.43        -0.38        0.16
  (3#lt20k) vs (1#20-500k) |      -0.17       0.14    -1.19    0.23        -0.44        0.11
(3#20-500k) vs (1#20-500k) |      -0.04       0.14    -0.28    0.78        -0.32        0.24
 (3#gt500k) vs (1#20-500k) |       0.03       0.15     0.21    0.83        -0.26        0.32
   (2#lt20k) vs (1#gt500k) |       0.05       0.14     0.38    0.70        -0.22        0.33
 (2#20-500k) vs (1#gt500k) |      -0.04       0.14    -0.26    0.79        -0.32        0.24
  (2#gt500k) vs (1#gt500k) |      -0.08       0.15    -0.55    0.58        -0.37        0.21
   (3#lt20k) vs (1#gt500k) |      -0.14       0.15    -0.94    0.35        -0.43        0.15
 (3#20-500k) vs (1#gt500k) |      -0.01       0.15    -0.08    0.93        -0.31        0.29
  (3#gt500k) vs (1#gt500k) |       0.06       0.16     0.38    0.70        -0.24        0.36
  (2#20-500k) vs (2#lt20k) |      -0.09       0.04    -2.41    0.02        -0.16       -0.02
   (2#gt500k) vs (2#lt20k) |      -0.13       0.05    -2.63    0.01        -0.23       -0.03
    (3#lt20k) vs (2#lt20k) |      -0.19       0.14    -1.40    0.16        -0.46        0.08
  (3#20-500k) vs (2#lt20k) |      -0.07       0.14    -0.46    0.64        -0.34        0.21
   (3#gt500k) vs (2#lt20k) |       0.01       0.14     0.05    0.96        -0.28        0.29
 (2#gt500k) vs (2#20-500k) |      -0.04       0.06    -0.72    0.47        -0.16        0.07
  (3#lt20k) vs (2#20-500k) |      -0.10       0.14    -0.73    0.47        -0.38        0.17
(3#20-500k) vs (2#20-500k) |       0.02       0.14     0.17    0.86        -0.26        0.31
 (3#gt500k) vs (2#20-500k) |       0.10       0.15     0.66    0.51        -0.19        0.39
   (3#lt20k) vs (2#gt500k) |      -0.06       0.14    -0.41    0.68        -0.34        0.22
 (3#20-500k) vs (2#gt500k) |       0.07       0.15     0.45    0.65        -0.22        0.36
  (3#gt500k) vs (2#gt500k) |       0.14       0.15     0.92    0.36        -0.16        0.44
  (3#20-500k) vs (3#lt20k) |       0.13       0.05     2.74    0.01         0.04        0.22
   (3#gt500k) vs (3#lt20k) |       0.20       0.07     2.88    0.00         0.06        0.33
 (3#gt500k) vs (3#20-500k) |       0.07       0.08     0.93    0.35        -0.08        0.22
----------------------------------------------------------------------------------------------
```

.19

2.2 Interactions with Financial Satisfaction

```
. mixed free i.t3##i.sf3    mar i.emp health kids rel_imp  || cc:   ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -184338.27
Iteration 1:  Log likelihood = -184338.27
Computing standard errors ...
Mixed-effects ML regression                     Number of obs    =   85,997
Group variable: cc                              Number of groups =       62
                                                Obs per group:
                                                              min =      849
                                                              avg =  1,387.0
                                                              max =    4,018
                                                Wald chi2(19)    = 10247.87
Log likelihood = -184338.27                     Prob > chi2      =   0.0000
------------------------------------------------------------------------------
        free | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
          t3 |
      20-500k |      -0.10       0.04    -2.80    0.01        -0.17       -0.03
```

```
       gt500k  |      -0.06       0.04    -1.56    0.12        -0.14        0.02
               |
           sf3 |
             2 |       0.67       0.03    21.86    0.00         0.61        0.73
             3 |       1.51       0.03    45.85    0.00         1.45        1.58
               |
        t3#sf3 |
    20-500k#2  |       0.20       0.04     4.69    0.00         0.12        0.28
    20-500k#3  |       0.14       0.05     3.08    0.00         0.05        0.23
     gt500k#2  |       0.15       0.05     3.15    0.00         0.06        0.24
     gt500k#3  |       0.16       0.05     3.16    0.00         0.06        0.26
               |
           mar |       0.02       0.02     1.24    0.21        -0.01        0.05
               |
           emp |
    Part time  |      -0.09       0.03    -3.38    0.00        -0.15       -0.04
Self employed  |      -0.01       0.02    -0.50    0.62        -0.06        0.03
      Retired  |      -0.05       0.02    -2.17    0.03        -0.10       -0.01
    Housewife  |      -0.21       0.02    -8.89    0.00        -0.26       -0.16
     Students  |      -0.21       0.03    -6.10    0.00        -0.27       -0.14
   Unemployed  |      -0.14       0.03    -4.83    0.00        -0.20       -0.08
        Other  |      -0.21       0.07    -3.06    0.00        -0.35       -0.08
               |
        health |       0.37       0.01    42.11    0.00         0.35        0.39
          kids |       0.03       0.01     4.45    0.00         0.01        0.04
       rel_imp |       0.04       0.01     4.11    0.00         0.02        0.05
         _cons |       4.89       0.08    59.41    0.00         4.73        5.05
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Identity                 |
                  var(_cons) |       0.26       0.05          0.18        0.37
-----------------------------+------------------------------------------------
               var(Residual) |       4.25       0.02          4.21        4.29
------------------------------------------------------------------------------
LR test vs. linear model: chibar2(01) = 4212.60       Prob >= chibar2 = 0.0000
. est sto a
. margins,  at(t3=(1 2 3)  sf=(1 2 3))
Predictive margins                                  Number of obs = 85,997
Expression: Linear prediction, fixed portion, predict()
1._at: t3  = 1
       sf3 = 1
2._at: t3  = 1
       sf3 = 2
3._at: t3  = 1
       sf3 = 3
4._at: t3  = 2
       sf3 = 1
5._at: t3  = 2
       sf3 = 2
6._at: t3  = 2
       sf3 = 3
7._at: t3  = 3
       sf3 = 1
8._at: t3  = 3
       sf3 = 2
9._at: t3  = 3
       sf3 = 3
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |       6.39       0.07    92.12    0.00         6.26        6.53
          2  |       7.06       0.07   104.88    0.00         6.93        7.19
          3  |       7.91       0.07   115.69    0.00         7.77        8.04
          4  |       6.29       0.07    90.53    0.00         6.16        6.43
          5  |       7.16       0.07   106.88    0.00         7.03        7.29
          6  |       7.95       0.07   116.96    0.00         7.81        8.08
          7  |       6.33       0.07    87.90    0.00         6.19        6.47
          8  |       7.15       0.07   104.65    0.00         7.01        7.28
          9  |       8.00       0.07   114.23    0.00         7.87        8.14
------------------------------------------------------------------------------
. marginsplot, xdimension(t3) xsize(2) ysize(2) saving(z1,replace) text(8.25 4 "{bf:a}", size(large))legend(off)text(7.87 1.32 "top 3 cat")text
> (7 1.32 "middle 3 cat")text(6.3 1.32 "bottom 4 cat")title("")ytitle("personal agency")scheme(s2mono)xsc(off)recast(line) //xtitle("rural-urba
> n")
Variables that uniquely identify margins: t3 sf3
file z1.gph saved
. mixed polSay i.t3##i.sf3   i.X025R  i.emp health  rel_imp  || cc:    ,mle //final! ditched mar and kids and added educ
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -103869.52
Iteration 1:  Log likelihood = -103869.52  (backed up)
Computing standard errors ...
Mixed-effects ML regression                     Number of obs    =  66,483
Group variable: cc                              Number of groups =      50
                                                Obs per group:
                                                              min =     416
                                                              avg = 1,329.7
                                                              max =   3,997
                                                Wald chi2(19)    =  862.78
Log likelihood = -103869.52                     Prob > chi2      =  0.0000
------------------------------------------------------------------------------
       polSay | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
--------------+---------------------------------------------------------------
           t3 |
      20-500k |      -0.04       0.02    -1.99    0.05        -0.09       -0.00
       gt500k |      -0.03       0.03    -1.17    0.24        -0.08        0.02
              |
          sf3 |
            2 |       0.10       0.02     5.22    0.00         0.06        0.14
            3 |       0.19       0.02     9.56    0.00         0.15        0.23
              |
       t3#sf3 |
    20-500k#2 |       0.04       0.03     1.35    0.18        -0.02        0.09
    20-500k#3 |       0.04       0.03     1.50    0.13        -0.01        0.10
```

```
     gt500k#2  |       0.02       0.03     0.80    0.42        -0.03        0.08
     gt500k#3  |       0.03       0.03     0.81    0.42        -0.04        0.09
               |
         X025R |
       Middle  |       0.00       0.01     0.03    0.98        -0.02        0.02
        Upper  |       0.04       0.01     3.42    0.00         0.02        0.07
               |
           emp |
    Part time  |      -0.00       0.02    -0.18    0.86        -0.04        0.03
Self employed  |      -0.02       0.01    -1.38    0.17        -0.05        0.01
      Retired  |       0.00       0.02     0.17    0.86        -0.03        0.03
    Housewife  |      -0.07       0.02    -4.36    0.00        -0.10       -0.04
     Students  |       0.00       0.02     0.24    0.81        -0.04        0.05
   Unemployed  |      -0.06       0.02    -3.22    0.00        -0.10       -0.02
        Other  |      -0.02       0.05    -0.49    0.62        -0.11        0.07
               |
       health  |       0.08       0.01    13.80    0.00         0.06        0.09
      rel_imp  |       0.07       0.01    12.79    0.00         0.06        0.08
        _cons  |       2.30       0.06    38.08    0.00         2.18        2.42
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Identity                 |
                  var(_cons) |       0.13       0.03          0.09        0.20
-----------------------------+------------------------------------------------
               var(Residual) |       1.33       0.01          1.31        1.34
------------------------------------------------------------------------------
LR test vs. linear model: chibar2(01) = 5171.45       Prob >= chibar2 = 0.0000
. est sto a
. margins,  at(t3=(1 2 3)  sf=(1 2 3))
Predictive margins                                 Number of obs = 66,483
Expression: Linear prediction, fixed portion, predict()
1._at: t3  = 1
       sf3 = 1
2._at: t3  = 1
       sf3 = 2
3._at: t3  = 1
       sf3 = 3
4._at: t3  = 2
       sf3 = 1
5._at: t3  = 2
       sf3 = 2
6._at: t3  = 2
       sf3 = 3
7._at: t3  = 3
       sf3 = 1
8._at: t3  = 3
       sf3 = 2
9._at: t3  = 3
       sf3 = 3
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |       2.80       0.05    52.26    0.00         2.70        2.91
          2  |       2.90       0.05    55.09    0.00         2.80        3.01
          3  |       3.00       0.05    56.44    0.00         2.89        3.10
          4  |       2.76       0.05    51.09    0.00         2.65        2.87
          5  |       2.89       0.05    54.94    0.00         2.79        3.00
          6  |       3.00       0.05    56.43    0.00         2.89        3.10
          7  |       2.77       0.06    50.20    0.00         2.67        2.88
          8  |       2.90       0.05    54.34    0.00         2.79        3.00
          9  |       2.99       0.05    55.34    0.00         2.89        3.10
------------------------------------------------------------------------------
. marginsplot ,xdimension(t3) xsize(2) ysize(2) saving(z2,replace) text(3.12 4 "{bf:b}", size(large))legend(off)text(2.98 1.42 "top 3 cat")text
> (2.88 1.42 "middle 3 cat")text(2.77 1.42 "bottom 4 cat")title("")ytitle("political efficacy")xtitle("rural-urban")scheme(s2mono)recast(line)
Variables that uniquely identify margins: t3 sf3
file z2.gph saved
. graph combine z1.gph z2.gph, cols(1) scheme(s2mono) xsize(2) ysize(4) scale(1)
. gr export zz.pdf,replace
file zz.pdf saved as PDF format
```

.19

2.2.1 which differences are significant at what level

`Personal Agency:` People with limited material resources report significantly more personal agency in rural areas than in town areas (b=0.10**); no other significant differences emerge. People with moderate material resources report more personal agency when residing in town areas (b=0.10***) and city areas (b=0.09***) as compared to rural areas, yet similar levels when comparing town and city areas. People with substantial material resources report significantly higher levels of personal agency in city areas than rural areas (b=0.10**); no other significant differences emerge.

`Political Agency:` With one exception, no significant urban-rural differences remain, once investigated seperately by individuls level of material resources - suggesting that much of the urban-rural-divide in political agency is due to differences in material resources. The only small difference that emerges indicates that people with limited resources in rural areas report slightly more political agency (b=0.04*) than in town areas

```
. //this is for free
```

FIRST JUST ONLY RELEVANT ONES, and below that all

	Contrast	Delta-method std. err.	z	P>|z|	Unadjusted [95% conf.	interval]
_at#t3						
(1#20-500k) vs (1#lt20k)	-0.10	0.04	-2.80	0.01	-0.17	-0.03
(1#gt500k) vs (1#lt20k)	-0.06	0.04	-1.56	0.12	-0.14	0.02
(1#gt500k) vs (1#20-500k)	0.04	0.04	0.88	0.38	-0.04	0.12
(2#20-500k) vs (2#lt20k)	0.10	0.03	3.77	0.00	0.05	0.15
(2#gt500k) vs (2#lt20k)	0.09	0.03	2.85	0.00	0.03	0.15
(2#gt500k) vs (2#20-500k)	-0.01	0.03	-0.47	0.64	-0.07	0.04
(3#20-500k) vs (3#lt20k)	0.04	0.03	1.29	0.20	-0.02	0.10
(3#gt500k) vs (3#lt20k)	0.10	0.04	2.77	0.01	0.03	0.17
(3#gt500k) vs (3#20-500k)	0.06	0.03	1.71	0.09	-0.01	0.13

```
. //this is for free
. margins t3, at(sf3=(1 2 3)) pwcompare(effects)
Pairwise comparisons of predictive margins              Number of obs = 85,997
Expression: Linear prediction, fixed portion, predict()
1._at: sf3 = 1
2._at: sf3 = 2
3._at: sf3 = 3
```

	Contrast	Delta-method std. err.	z	P>|z|	Unadjusted [95% conf.	interval]
_at#t3						
(1#20-500k) vs (1#lt20k)	-0.10	0.04	-2.80	0.01	-0.17	-0.03
(1#gt500k) vs (1#lt20k)	-0.06	0.04	-1.56	0.12	-0.14	0.02
(2#lt20k) vs (1#lt20k)	0.67	0.03	21.86	0.00	0.61	0.73
(2#20-500k) vs (1#lt20k)	0.77	0.03	24.51	0.00	0.71	0.83
(2#gt500k) vs (1#lt20k)	0.76	0.03	21.82	0.00	0.69	0.82
(3#lt20k) vs (1#lt20k)	1.51	0.03	45.85	0.00	1.45	1.58
(3#20-500k) vs (1#lt20k)	1.55	0.03	46.14	0.00	1.49	1.62
(3#gt500k) vs (1#lt20k)	1.61	0.04	42.28	0.00	1.54	1.69
(1#gt500k) vs (1#20-500k)	0.04	0.04	0.88	0.38	-0.04	0.12
(2#lt20k) vs (1#20-500k)	0.77	0.03	23.92	0.00	0.71	0.83
(2#20-500k) vs (1#20-500k)	0.87	0.03	28.57	0.00	0.81	0.93
(2#gt500k) vs (1#20-500k)	0.86	0.03	24.99	0.00	0.79	0.92
(3#lt20k) vs (1#20-500k)	1.61	0.03	46.59	0.00	1.55	1.68
(3#20-500k) vs (1#20-500k)	1.65	0.03	50.32	0.00	1.59	1.72
(3#gt500k) vs (1#20-500k)	1.71	0.04	45.24	0.00	1.64	1.79
(2#lt20k) vs (1#gt500k)	0.73	0.04	19.50	0.00	0.66	0.81
(2#20-500k) vs (1#gt500k)	0.83	0.04	22.78	0.00	0.76	0.91
(2#gt500k) vs (1#gt500k)	0.82	0.04	22.23	0.00	0.75	0.89
(3#lt20k) vs (1#gt500k)	1.58	0.04	39.69	0.00	1.50	1.66
(3#20-500k) vs (1#gt500k)	1.62	0.04	41.79	0.00	1.54	1.69
(3#gt500k) vs (1#gt500k)	1.68	0.04	41.27	0.00	1.60	1.76
(2#20-500k) vs (2#lt20k)	0.10	0.03	3.77	0.00	0.05	0.15
(2#gt500k) vs (2#lt20k)	0.09	0.03	2.85	0.00	0.03	0.15
(3#lt20k) vs (2#lt20k)	0.84	0.03	30.44	0.00	0.79	0.90
(3#20-500k) vs (2#lt20k)	0.88	0.03	30.71	0.00	0.83	0.94
(3#gt500k) vs (2#lt20k)	0.94	0.03	27.85	0.00	0.88	1.01
(2#gt500k) vs (2#20-500k)	-0.01	0.03	-0.47	0.64	-0.07	0.04
(3#lt20k) vs (2#20-500k)	0.74	0.03	25.59	0.00	0.69	0.80
(3#20-500k) vs (2#20-500k)	0.78	0.03	29.99	0.00	0.73	0.84
(3#gt500k) vs (2#20-500k)	0.84	0.03	25.80	0.00	0.78	0.91
(3#lt20k) vs (2#gt500k)	0.76	0.03	23.33	0.00	0.69	0.82
(3#20-500k) vs (2#gt500k)	0.80	0.03	25.70	0.00	0.74	0.86
(3#gt500k) vs (2#gt500k)	0.86	0.03	26.32	0.00	0.79	0.92
(3#20-500k) vs (3#lt20k)	0.04	0.03	1.29	0.20	-0.02	0.10
(3#gt500k) vs (3#lt20k)	0.10	0.04	2.77	0.01	0.03	0.17
(3#gt500k) vs (3#20-500k)	0.06	0.03	1.71	0.09	-0.01	0.13

```
//this is for polSay
FIRST JUST ONLY RELEVANT ONES, and below that all
```

	Contrast	Delta-method std. err.	z	P>|z|	Unadjusted [95% conf.	interval]
_at#t3						
(1#20-500k) vs (1#lt20k)	-0.04	0.02	-1.99	0.05	-0.09	-0.00
(1#gt500k) vs (1#lt20k)	-0.03	0.03	-1.17	0.24	-0.08	0.02
(1#gt500k) vs (1#20-500k)	0.01	0.03	0.54	0.59	-0.04	0.07
(2#20-500k) vs (2#lt20k)	-0.01	0.02	-0.45	0.65	-0.04	0.03
(2#gt500k) vs (2#lt20k)	-0.01	0.02	-0.33	0.74	-0.04	0.03
(2#gt500k) vs (2#20-500k)	0.00	0.02	0.07	0.94	-0.04	0.04
(3#20-500k) vs (3#lt20k)	-0.00	0.02	-0.09	0.93	-0.04	0.04
(3#gt500k) vs (3#lt20k)	-0.00	0.02	-0.20	0.84	-0.05	0.04
(3#gt500k) vs (3#20-500k)	-0.00	0.02	-0.12	0.90	-0.05	0.04

```
. //this is for polSay
. margins t3, at(sf3=(1 2 3)) pwcompare(effects)
Pairwise comparisons of predictive margins              Number of obs = 66,483
Expression: Linear prediction, fixed portion, predict()
1._at: sf3 = 1
2._at: sf3 = 2
3._at: sf3 = 3
```

	Contrast	Delta-method std. err.	z	P>|z|	Unadjusted [95% conf.	interval]
_at#t3						
(1#20-500k) vs (1#lt20k)	-0.04	0.02	-1.99	0.05	-0.09	-0.00
(1#gt500k) vs (1#lt20k)	-0.03	0.03	-1.17	0.24	-0.08	0.02
(2#lt20k) vs (1#lt20k)	0.10	0.02	5.22	0.00	0.06	0.14
(2#20-500k) vs (1#lt20k)	0.09	0.02	4.60	0.00	0.05	0.13
(2#gt500k) vs (1#lt20k)	0.09	0.02	4.22	0.00	0.05	0.14
(3#lt20k) vs (1#lt20k)	0.19	0.02	9.56	0.00	0.15	0.23
(3#20-500k) vs (1#lt20k)	0.19	0.02	9.09	0.00	0.15	0.23

```
   (3#gt500k) vs (1#lt20k)  |       0.19       0.02     7.93    0.00        0.14        0.24
 (1#gt500k) vs (1#20-500k)  |       0.01       0.03     0.54    0.59       -0.04        0.07
  (2#lt20k) vs (1#20-500k)  |       0.14       0.02     6.96    0.00        0.10        0.18
(2#20-500k) vs (1#20-500k)  |       0.14       0.02     6.70    0.00        0.10        0.18
 (2#gt500k) vs (1#20-500k)  |       0.14       0.02     6.14    0.00        0.09        0.18
  (3#lt20k) vs (1#20-500k)  |       0.24       0.02    10.86    0.00        0.19        0.28
(3#20-500k) vs (1#20-500k)  |       0.24       0.02    10.97    0.00        0.19        0.28
 (3#gt500k) vs (1#20-500k)  |       0.23       0.02     9.62    0.00        0.19        0.28
   (2#lt20k) vs (1#gt500k)  |       0.13       0.02     5.37    0.00        0.08        0.18
 (2#20-500k) vs (1#gt500k)  |       0.12       0.02     5.10    0.00        0.07        0.17
  (2#gt500k) vs (1#gt500k)  |       0.12       0.02     5.22    0.00        0.08        0.17
   (3#lt20k) vs (1#gt500k)  |       0.22       0.03     8.91    0.00        0.17        0.27
 (3#20-500k) vs (1#gt500k)  |       0.22       0.02     8.93    0.00        0.17        0.27
  (3#gt500k) vs (1#gt500k)  |       0.22       0.03     8.56    0.00        0.17        0.27
  (2#20-500k) vs (2#lt20k)  |      -0.01       0.02    -0.45    0.65       -0.04        0.03
   (2#gt500k) vs (2#lt20k)  |      -0.01       0.02    -0.33    0.74       -0.04        0.03
    (3#lt20k) vs (2#lt20k)  |       0.09       0.02     5.55    0.00        0.06        0.13
  (3#20-500k) vs (2#lt20k)  |       0.09       0.02     5.10    0.00        0.06        0.13
   (3#gt500k) vs (2#lt20k)  |       0.09       0.02     4.24    0.00        0.05        0.13
 (2#gt500k) vs (2#20-500k)  |       0.00       0.02     0.07    0.94       -0.04        0.04
  (3#lt20k) vs (2#20-500k)  |       0.10       0.02     5.59    0.00        0.07        0.14
(3#20-500k) vs (2#20-500k)  |       0.10       0.02     5.82    0.00        0.07        0.13
 (3#gt500k) vs (2#20-500k)  |       0.10       0.02     4.67    0.00        0.06        0.14
   (3#lt20k) vs (2#gt500k)  |       0.10       0.02     4.94    0.00        0.06        0.14
 (3#20-500k) vs (2#gt500k)  |       0.10       0.02     4.97    0.00        0.06        0.14
  (3#gt500k) vs (2#gt500k)  |       0.10       0.02     4.76    0.00        0.06        0.14
  (3#20-500k) vs (3#lt20k)  |      -0.00       0.02    -0.09    0.93       -0.04        0.04
   (3#gt500k) vs (3#lt20k)  |      -0.00       0.02    -0.20    0.84       -0.05        0.04
 (3#gt500k) vs (3#20-500k)  |      -0.00       0.02    -0.12    0.90       -0.05        0.04
----------------------------------------------------------------------------------------------
.
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.19